

RDFLib: Python based library for the semantic Web

RDFLib is a powerful Python package for RDF semantic Web programming. It contains the standards required to work with RDF. It includes the parsers and serialisers for RDF/XML, NTriples, N-Quads, N3, RDFa, Turtle, TriX, Microdata and many others. RDFLib has the Graph interface so that the creation of connections and visualisation is quite easy to implement.

The features of RDFLib are:

- Loads and saves RDF
- Creates RDF triples and relationships
- Navigates graphs
- Merges graphs
- Name spaces
- Bindings
- Interconnections
- Querying with SPARQL
- Utilities and convenience functions

Installing RDFLib with Python

On Anaconda Prompt, the following instruction can be executed for the installation of RDFLib:

```
Anaconda Prompt > pip install rdflib
```

After this, the RDFLib functions will be available for programming with the semantic Web.

```
Python Programming with RDF
import rdflib
g=rdflib.Graph()
g.load('http://dbpedia.org/resource/Semantic_Web')
for s,p,o in g:
    print (s,p,o)
```

Querying with SPARQL

The SPARQL query is used for the mapping of a FOAF file with the Python code so that the dynamic extraction of data from the FOAF RDF file can be done. Here, the FOAF RDF file *myfoaf.rdf* is used as the back-end database. To query this database, the SPARQL query

```
g.V().has("name","gremlin").
    out("knows").
    out("knows").
    values("name")
```

What are the names of Gremlin's friends' friends?

1. Get the vertex with name "gremlin."
2. Traverse to the people that Gremlin knows.
3. Traverse to the people those people know.
4. Get those people's names.

Figure 3: Extraction of knowledge and relationships of a person called Gremlin

```
g.V().match(
    as("a").out("knows").as("b"),
    as("a").out("created").as("c"),
    as("b").out("created").as("c"),
    as("c").in("created").count().is(2)).
    select("c").by("name")
```

What are the names of the projects created by two friends?

1. ...there exists some "a" who knows "b".
2. ...there exists some "a" who created "c".
3. ...there exists some "b" who created "c".
4. ...there exists some "c" created by 2 people.
5. Get the name of all matching "c" projects.

Figure 4: Extraction of projects created by the person

يزداد ورود المعلومات، تزداد مقدرات معالجة المعلومات أوتوماتيكياً، لذا يوجد إمكانيات كبيرة للاستفادة من مقدرات الأتمتة هذه بهدف استخراج المعلومات والخدمات من فيضان الويب والمرتبطة بالمستخدم، وتوصيلها إليه عن طريق واجهة مستخدم معيارية (Standardized User Interface)
http://dbpedia.org/resource/Semantic_Web <http://www.w3.org/2002/07/owl#sameAs> <http://www.wikidata.org/entity/Q54837>

Figure 5: Fetching results about a specific subject using RDFLib

is fired rather than SQL.

```
import rdflib
mygraph = rdflib.Graph()
mygraph.parse("myfoaf.rdf")
mygraph qres = mygraph.query(
    """SELECT DISTINCT ?fname ?lname
    WHERE {
        ?a foaf:knows ?b .
        ?a foaf:name ?fname .
        ?b foaf:name ?lname .
    }""")
for myrow in qres:
    print("%s knows %s" % row)
```

In traditional RDBMS databases, an SQL query is used for query processing. In case of semantic Web applications, the SPARQL query

extracts the information associated from the ontology file and FOAF association data set. This type of data extraction is used to extract hidden associations for digital forensic applications.

Scope of research and development

Semantic Web programming is typically dependent on the associations and relationships mentioned in RDF, FOAF and OWL. These knowledge representation standards and protocols can be enhanced so that hidden fingerprints in the data items and their relationships can be associated. Using this approach, indirect relationships about a specific person or object in the database can be extracted. **END** 

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